

Final Insights & Recommendations

ML Engine — End-to-End · Generated: 2026-05-23 15:14

1. Dataset Summary

Item	Value
File	superstore_clean.csv
Shape	9,994 rows × 20 cols
Target	Profit
Features selected	1

2. EDA Insights (Stage 1)

INSIGHTS & RECOMMENDATIONS REPORT

Dataset : superstore_clean.csv

Target : Profit

Shape : 9,994 rows × 20 columns

1. CORRELATION FINDINGS

Variables most correlated with 'Profit':

- Sales $r = +0.4791$ (positive)
- is_profitable $r = +0.2287$ (positive)
- profit_margin $r = +0.2240$ (positive)
- Discount $r = -0.2195$ (negative)
- is_discounted $r = -0.1569$ (negative)

KEY INSIGHT: 'Sales' shows the strongest correlation ($r = +0.4791$) with Profit.

2. DATA QUALITY ASSESSMENT

Total missing cells : 0

■ No missing values detected.

3. OUTLIER SUMMARY (IQR method, 1.5×)

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- Sales 1,167 outliers (11.7%) [SEVERE]
- Quantity 170 outliers (1.7%) [CLEAN]
- Discount 856 outliers (8.6%) [MODERATE]
- Profit 1,881 outliers (18.8%) [SEVERE]
- delivery_days 0 outliers (0.0%) [CLEAN]
- order_year 0 outliers (0.0%) [CLEAN]
- profit_margin 952 outliers (9.5%) [MODERATE]

- is_profitable 1,936 outliers (19.4%) [SEVERE]
- is_discounted 0 outliers (0.0%) [CLEAN]

4. SKEWNESS ASSESSMENT



Highly skewed variables ($|skew| > 1$):

- Sales skewness = 12.973
- Profit skewness = 7.561
- profit_margin skewness = -2.893
- Discount skewness = 1.684
- is_profitable skewness = -1.550
- Quantity skewness = 1.279

RECOMMENDATION: Apply log/sqrt transformation to these variables before modelling.

5. MODELLING RECOMMENDATIONS



- Use the 1 variables with $corr \geq 0.3$ as features.
- Run VIF check (Tab 7) to detect multicollinearity before Regression.
- Apply IQR cleaning (Tab 3) to remove or cap extreme outliers.
- Handle any missing values (Tab 6) before splitting train/test.

3. Regression Model Results (Stage 2)

Model	R ²	MAE	RMSE	CV R ²
Random Forest	0.1337	11	205	0.9170±0.0771
Gradient Boosting	-0.0319	12	224	0.9306±0.0537
Decision Tree	-0.3805	24	259	0.8277±0.0635
Lasso Regression	-0.7113	64	288	0.3904±0.1653
Ridge Regression	-0.7131	65	288	0.3903±0.1655
Linear Regression	-0.7133	65	288	0.3902±0.1656

■ Best Regression Model: **Random Forest** R² = 0.1337

4. Classification Model Results (Stage 2)

Model	Accuracy	F1	Precision	Recall	CV Acc
Decision Tree	100.00%	1.0000	1.0000	1.0000	100.0%
K-Nearest Neighbors	100.00%	1.0000	1.0000	1.0000	100.0%

Random Forest	100.00%	1.0000	1.0000	1.0000	100.0%
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■ Best Classification Model: **Decision Tree** Accuracy = 100.00%

5. Final Recommendations

- Verify missing value imputation (Tab 6)
- Check VIF multicollinearity (Tab 7)
- Review outlier treatment (Tab 3)
- Use Random Forest for regression tasks
- Use Decision Tree for classification tasks
- Run Predict New Data (Tab 12) for inference

Report generated by ML Engine — Stage 1 + Stage 2 · Mohamed